Analysis of Monolithic and Microservices Architecture

Design, Scalability, and Trade-offs

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INTRODUCTION

Two Architectural Paradigms

Monolithic Architecture

- Monolithic architecture is a traditional software design approach where the entire application is built as a single unified system.
- All functional components such as user interface, business logic, and data access are tightly integrated in one codebase.
- The application is developed, deployed, and scaled as one unit.
- Components share the same database and runtime environment.
- It is widely used in small to medium-scale applications due to its simplicity.

Microservices Architecture

- Microservices architecture is a modern approach where an application is divided into multiple small, independent services.
- Each service represents a specific business capability and runs in its own process.
- Services communicate with each other through APIs or messaging.
- Each microservice can be developed, deployed, and scaled independently.
- It is widely adopted in large-scale and cloud-native systems.

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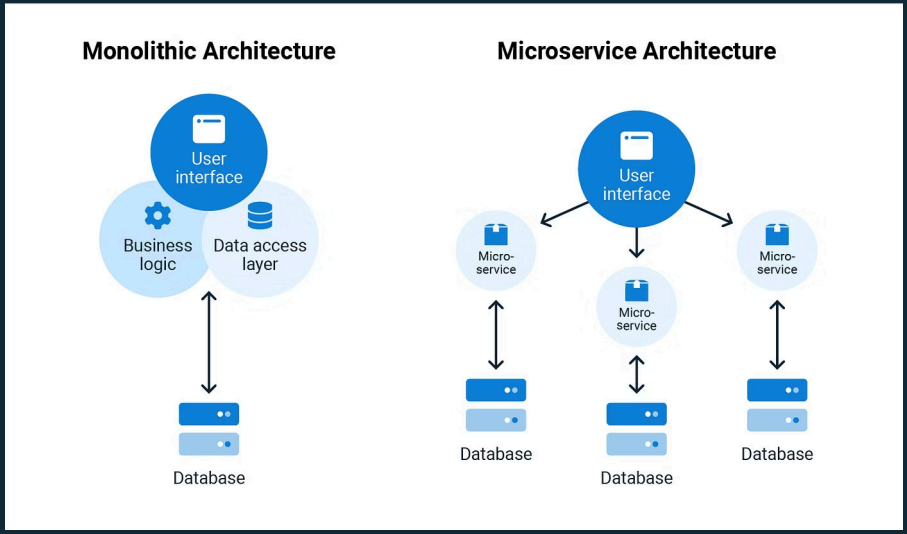
ARCHITECTURE FLOW

Architecture and Working Flow



Monolithic Flow

- Represented as a single system boundary containing all application components.
- Internal modules (UI, business logic, data access) are modeled as tightly coupled components.
- Components interact through direct associations within the same deployment unit.
- A shared database is depicted as a common dependency for all modules.
- Emphasizes centralized control and in-process communication inside one application container.

Microservices Flow

- Modeled as multiple independent service components, each with its own system boundary.
- Every service is associated with a dedicated datastore, indicating decentralized data ownership.
- External clients interact through an API Gateway or service interface component.
- Inter-service relationships are shown via interface-based or message-based communication links.
- Highlights distributed structure, loose coupling, and independent deployment of services.

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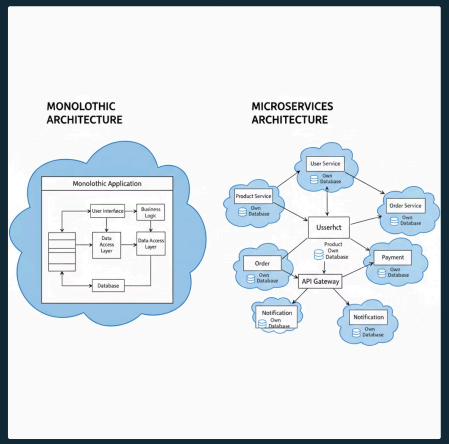
UML OVERVIEW

UML Diagram Comparison



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COMPARISON

Pros and Cons Analysis

Monolithic Architecture

Pros

- Simple architecture and straightforward development
- Easy testing and debugging within single codebase
- Lower initial deployment complexity
- Efficient in-process communication
- Ideal for small to medium applications

Cons

- Limited scalability—entire app scales together
- Tight coupling hinders maintenance
- Technology stack lock-in
- Large codebase reduces agility
- Single point of failure

Microservices Architecture

Pros

- Independent service scalability
- Loose coupling improves flexibility
- Technology heterogeneity enabled
- Faster feature delivery
- High resilience and fault isolation

Cons

- Higher architectural complexity
- Requires advanced DevOps and monitoring
- Network latency overhead
- Distributed data management challenges
- Complex testing across services

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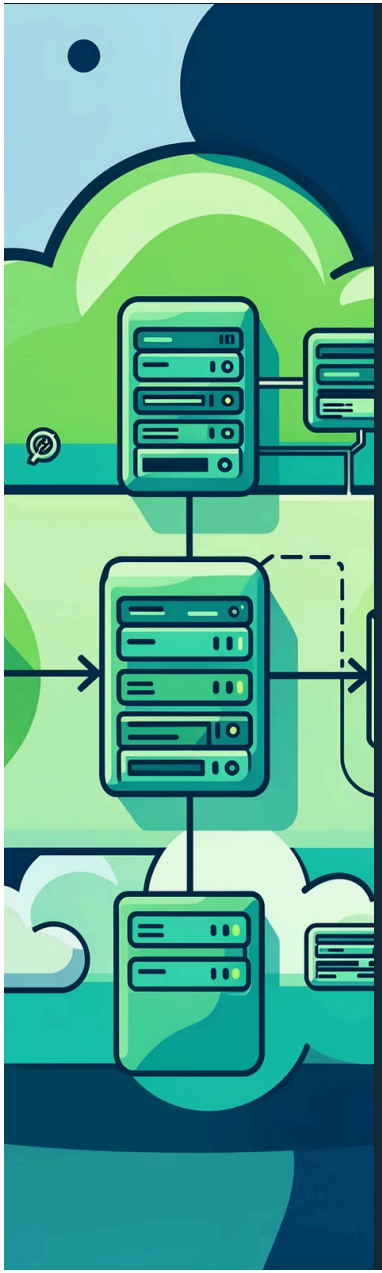


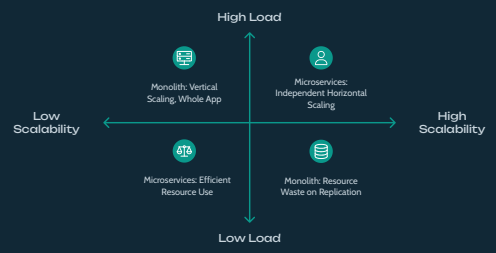
Scalability Analysis

Monolithic Scalability

- The whole application must be scaled together.
- If one module gets heavy traffic, entire system is replicated.
- Usually scaled by increasing server power or adding full copies.
- This wastes resources because all parts scale even if not needed.
- Scaling becomes difficult as the application grows large.

Microservices Scalability

- Only the required service is scaled based on demand.
- High-load services can scale without affecting others.
- New service instances can be added easily in cloud environments.
- Resources are used efficiently because scaling is targeted.
- Supports large-scale and high-traffic applications effectively.



Microservices architecture enables granular, targeted scaling that optimizes infrastructure costs while maintaining performance under variable load conditions.

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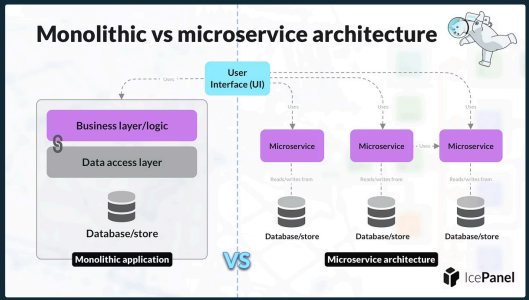
MIGRATION STORY

Why Applications Move from Monolithic to Microservices



- In monolithic systems, all features are combined in one application.
- As the system grows, it becomes large and difficult to manage.
- The whole application must be scaled even if only one part needs it.
- Any small change requires redeploying the entire system.
- Microservices separate features into independent services for better scalability and maintenance.

Example: Netflix — Monolithic to Microservices

- Netflix initially started with a monolithic architecture where streaming, user accounts, recommendations, and billing were part of one system.
- As global users increased, streaming and recommendation features required much higher scalability than other modules.
- Scaling the entire monolithic system became inefficient and difficult to maintain.
- Netflix migrated to microservices by separating services like Streaming Service, User Service, Recommendation Service, and Billing Service.

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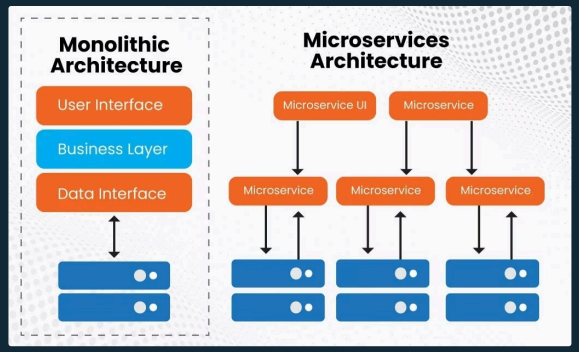
Market Analysis and Trade-offs



Market Trends

- Small and medium companies prefer monolithic for faster development and lower costs
- Startups begin with monolithic to build MVP products quickly
- Large enterprises adopt microservices for scalability and reliability
- Cloud computing and DevOps accelerate microservices adoption
- Modern platforms (Netflix, Amazon, Uber) leverage microservices at scale

Trade-off in Market

- Monolithic offers lower initial cost and simplicity, but limited long-term scalability.
- Microservices provide scalability and flexibility, but require higher infrastructure and expertise.
- Organizations choose architecture based on product size, team maturity, and growth expectations.
- Early-stage products favor monolithic, while scaling businesses transition to microservices.

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Conclusion

- Monolithic architecture provides simplicity, centralized control, and faster initial development.
- Microservices architecture enables scalability, flexibility, and independent service evolution.
- Monolithic systems are suitable for small to medium or stable applications.
- Microservices are better suited for large-scale, high-traffic, and continuously evolving systems.
- The choice of architecture depends on business size, growth expectations, and technical maturity.
- Industry trends show gradual migration from monolithic to microservices as systems scale.

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Thank You !!!

Questions and Discussion